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AWS
Organizations



AWS
Control Tower



AWS
IAM



AWS
Config



Amazon
CloudWatch



AWS
Trusted Advisor

AWS SOLUTIONS ARCHITECT ASSOCIATE

Instance Types, High Availability,
Fault Tolerance & Scaling Concepts.



GOVERN

Establish Strong
Cloud Governance
Frameworks



OPTIMIZE

Drive Efficiency
Through Visibility
and Control



SECURE

Enforce Security
and Compliance
at Scale



INNOVATE

Enable Resilient,
Scalable & Future-
Ready Architectures

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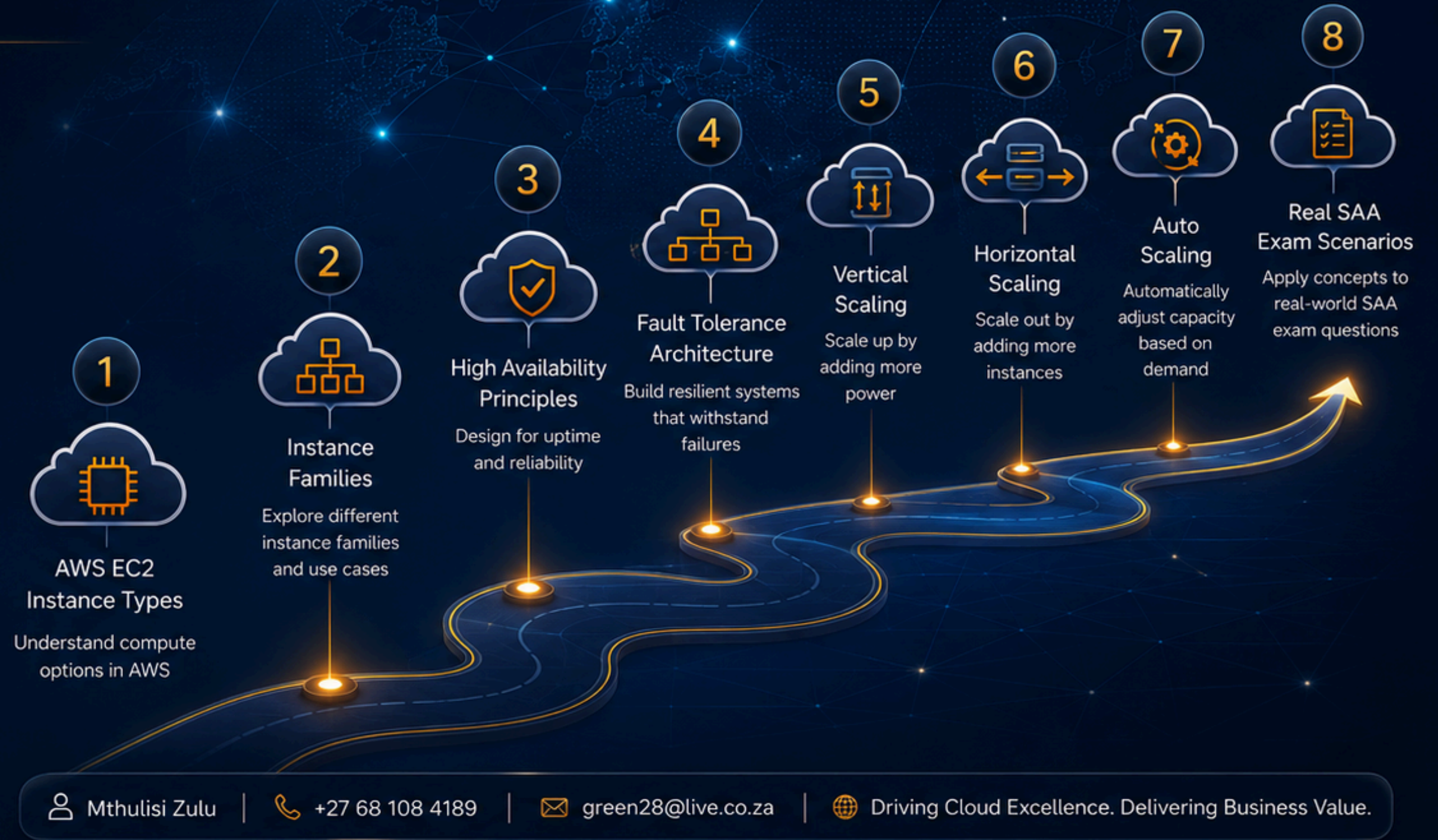
BUILDING AFRICA'S NEXT GENERATION OF CLOUD ARCHITECTS

TODAY'S LEARNING JOURNEY



By the end of this session students will understand:

- ✓ AWS EC2 Instance Types
- ✓ Instance Families
- ✓ High Availability Principles
- ✓ Fault Tolerance Architecture
- ✓ Vertical Scaling
- ✓ Horizontal Scaling
- ✓ Auto Scaling
- ✓ Real SAA Exam Scenarios



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BUILDING AFRICA'S NEXT GENERATION OF CLOUD ARCHITECTS

UNDERSTANDING EC2 INSTANCE TYPES

Different workloads require different compute resources.



Choosing the right instance type improves performance, optimizes costs, and drives architectural excellence.



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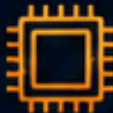
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INSTANCE SELECTION STRATEGY



Match the right EC2 instance family to your workload requirements.

WORKLOAD MAPPING TABLE

WORKLOAD	RECOMMENDED INSTANCE FAMILY
 Web Applications →  General Purpose (t3, t4g, m5)	
 Batch Processing →  Compute Optimized (c5, c6g)	
 SAP HANA →  Memory Optimized (r5, x1)	
 Big Data →  Storage Optimized (i3, d2)	



EXAM TIP

The exam often asks:

"Which instance family best suits this workload?"



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HIGH AVAILABILITY ARCHITECTURE



Ensures applications remain available despite failures.

AWS SERVICES



AVAILABILITY ZONES

Isolated locations within an AWS Region



ELASTIC LOAD BALANCER

Distributes incoming traffic across multiple targets



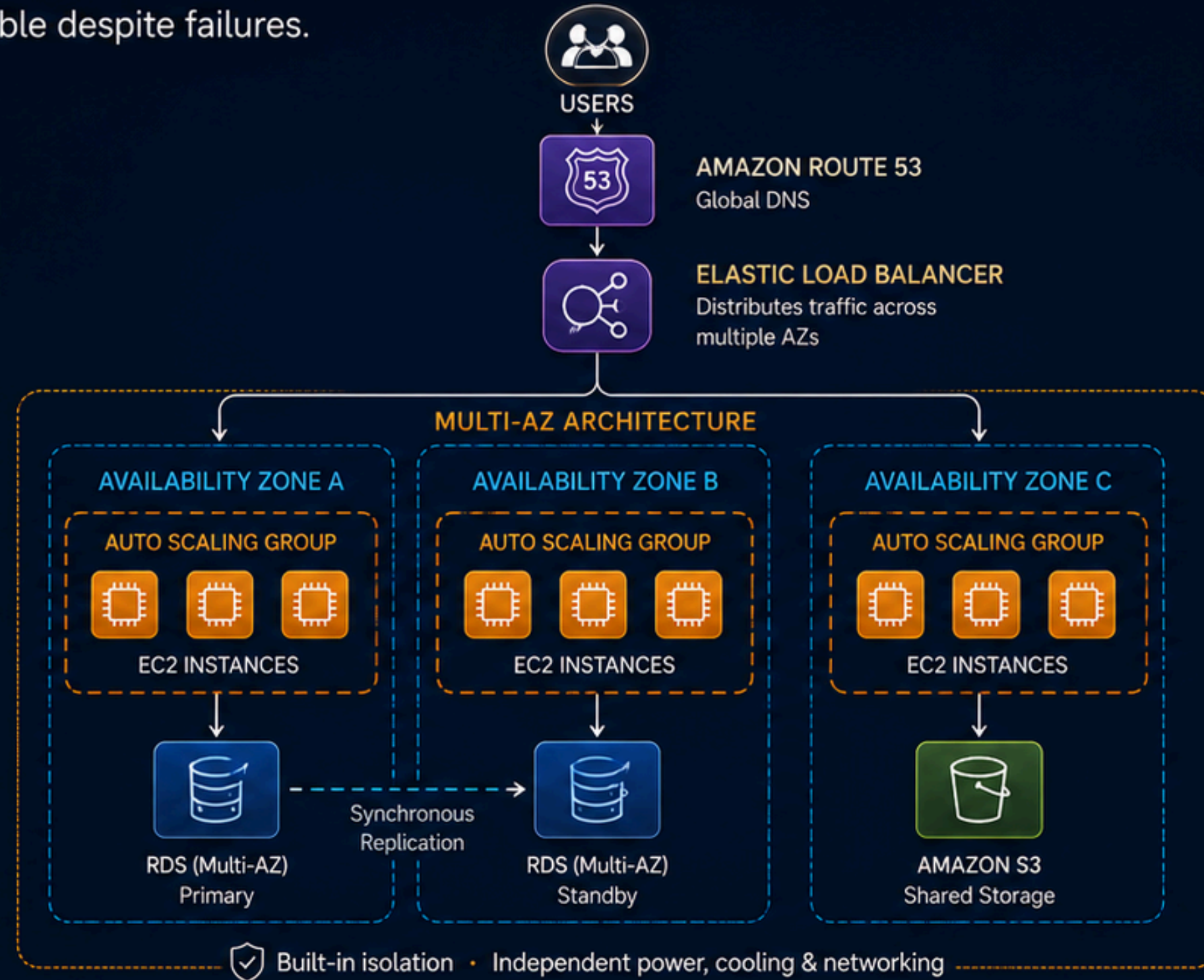
ROUTE 53

Global DNS service that routes users to your app



AUTO SCALING GROUPS

Automatically adjusts capacity to maintain availability



EXAMPLE

Application deployed across:

- ✓ AZ-A
- ✓ AZ-B
- ✓ AZ-C

BENEFITS

- ✓ Reduced downtime
- ✓ Improved reliability
- ✓ Better customer experience

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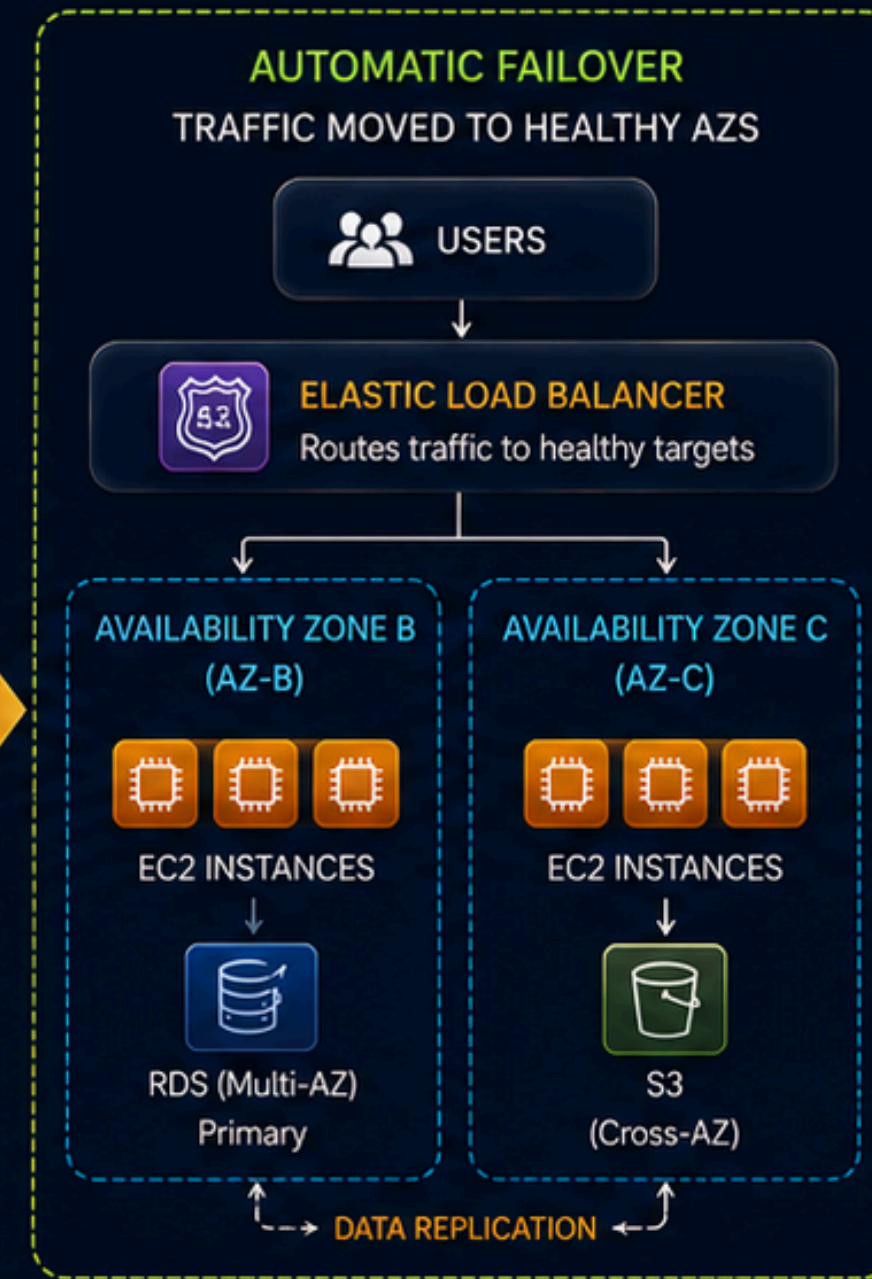
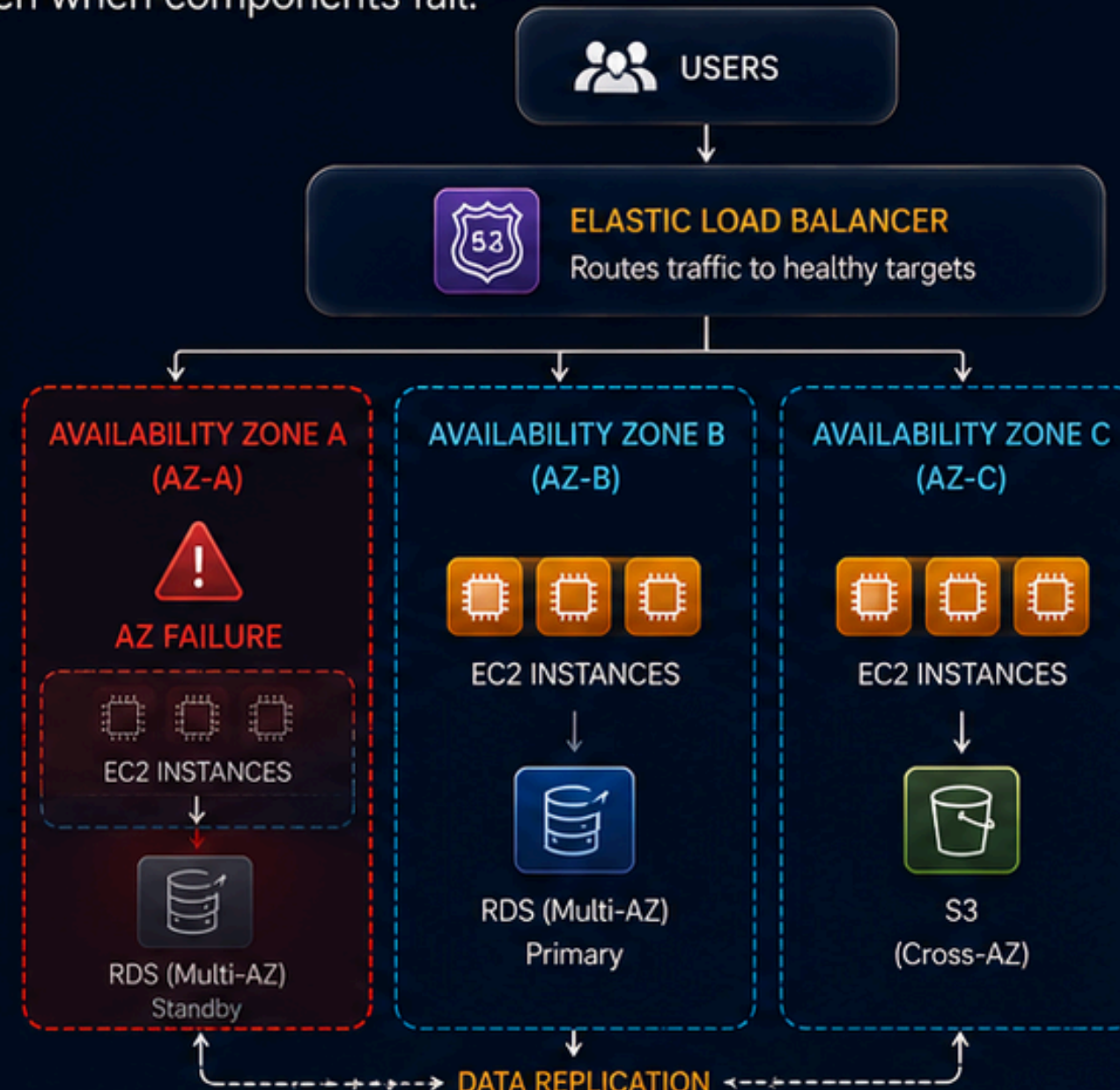
DESIGNING FOR FAILURE



Ability to continue operating even when components fail.

AWS EXAMPLES

-  **MULTI-AZ RDS**
Automatic failover for databases
-  **CROSS-AZ LOAD BALANCING**
Distributes traffic across healthy AZs
-  **MULTI-REGION ARCHITECTURES**
Replicate across regions for disaster resilience
-  **AUTO SCALING**
Maintains availability by adjusting capacity



SCENARIO

 **IF AZ-A FAILS:**
Traffic automatically moves to AZ-B

KEY PRINCIPLE

 **ASSUME EVERYTHING CAN FAIL.**
DESIGN ACCORDINGLY.

FAULT TOLERANCE BENEFITS



Minimizes downtime



Improves reliability



Enhances customer experience



Reduces operational risk & cost

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SCALING FUNDAMENTALS



Choose the right scaling approach based on your application needs.

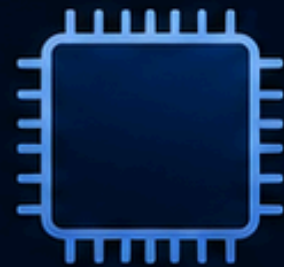
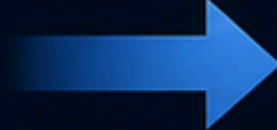
VERTICAL SCALING (SCALE UP)

Increase the size of an existing instance.



t3.micro

1 vCPU | 1 GB RAM



t3.large

2 vCPU | 8 GB RAM

EXAMPLE:

t3.micro → t3.large



Simple to implement



Limited by instance maximum size

VS

HORIZONTAL SCALING (SCALE OUT)

Increase the number of instances.



2 EC2 Instances



10 EC2 Instances

EXAMPLE:

2 EC2 → 10 EC2



Highly available



Fault tolerant



Cost effective with scale



SAA EXAM TIP

AWS prefers HORIZONTAL SCALING whenever possible.



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AUTOMATIC CAPACITY MANAGEMENT



Auto Scaling ensures the right number of instances are available to handle demand.

WHAT AUTO SCALING DOES

Automatically:

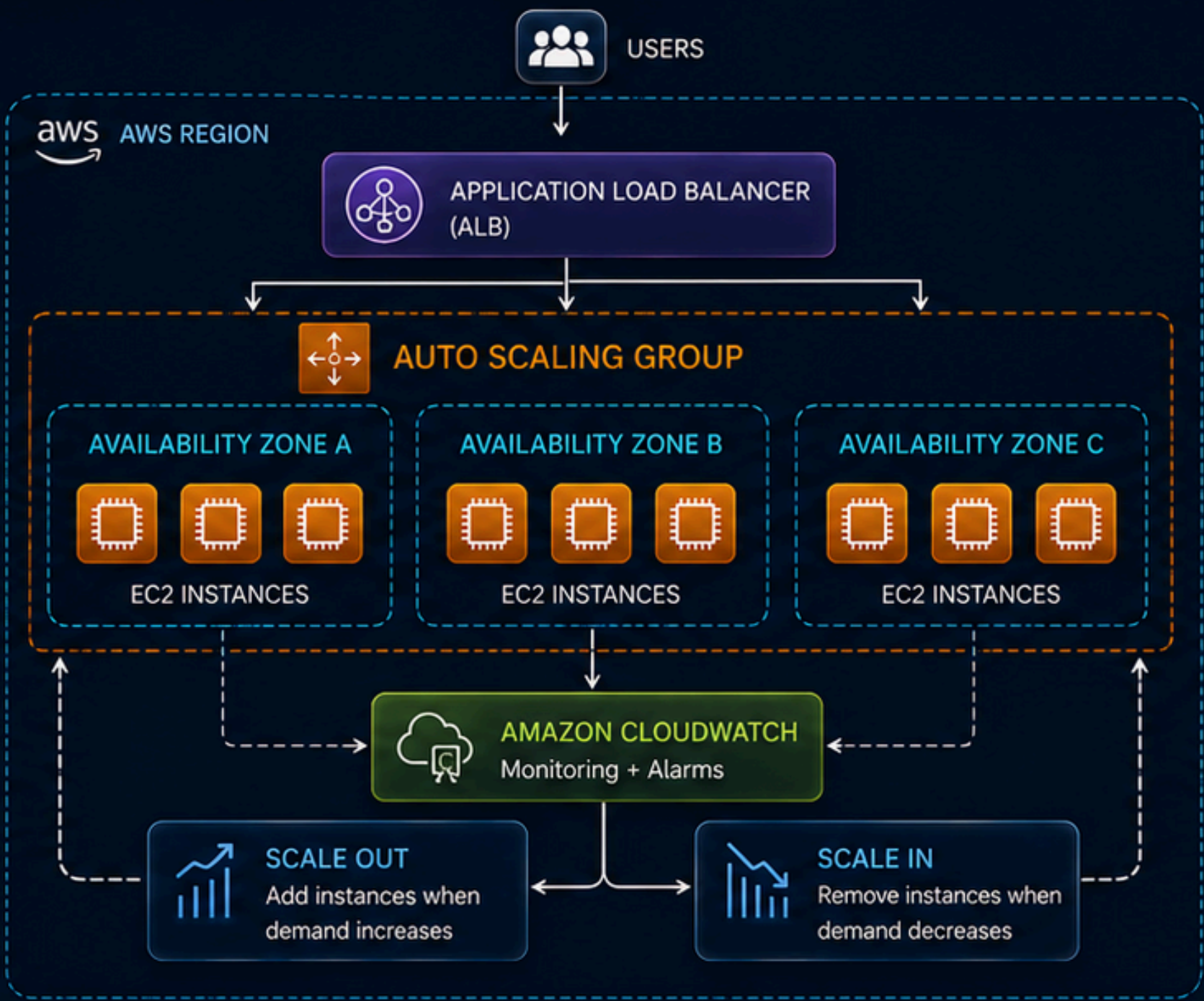
- ✓ Launches instances
- ✓ Terminates instances
- ✓ Maintains desired capacity

COMPONENTS

LOAD BALANCER
Distributes incoming traffic

AUTO SCALING GROUP
Manages EC2 instance capacity

EC2 FLEET
Launches and terminates instances as needed



BENEFITS

HIGH AVAILABILITY
Maintains application availability across multiple AZs

COST OPTIMIZATION
Scales in during low demand to reduce costs

ELASTICITY
Scales out quickly to handle traffic spikes

DEMAND VS CAPACITY

The graph shows **Demand** (blue line) fluctuating over **TIME**. **Capacity** (green line) follows the demand, scaling up when demand increases and scaling down when demand decreases. A dashed orange line represents the **Desired Capacity**.



AWS SAA SCENARIO ANALYSIS



Let's think like an Architect. What's the best solution?



SCENARIO

An e-commerce website experiences heavy traffic spikes during Black Friday.



REQUIREMENTS:

- ✓ Remain highly available
- ✓ Handle sudden growth
- ✓ Minimize costs



BEST SOLUTION



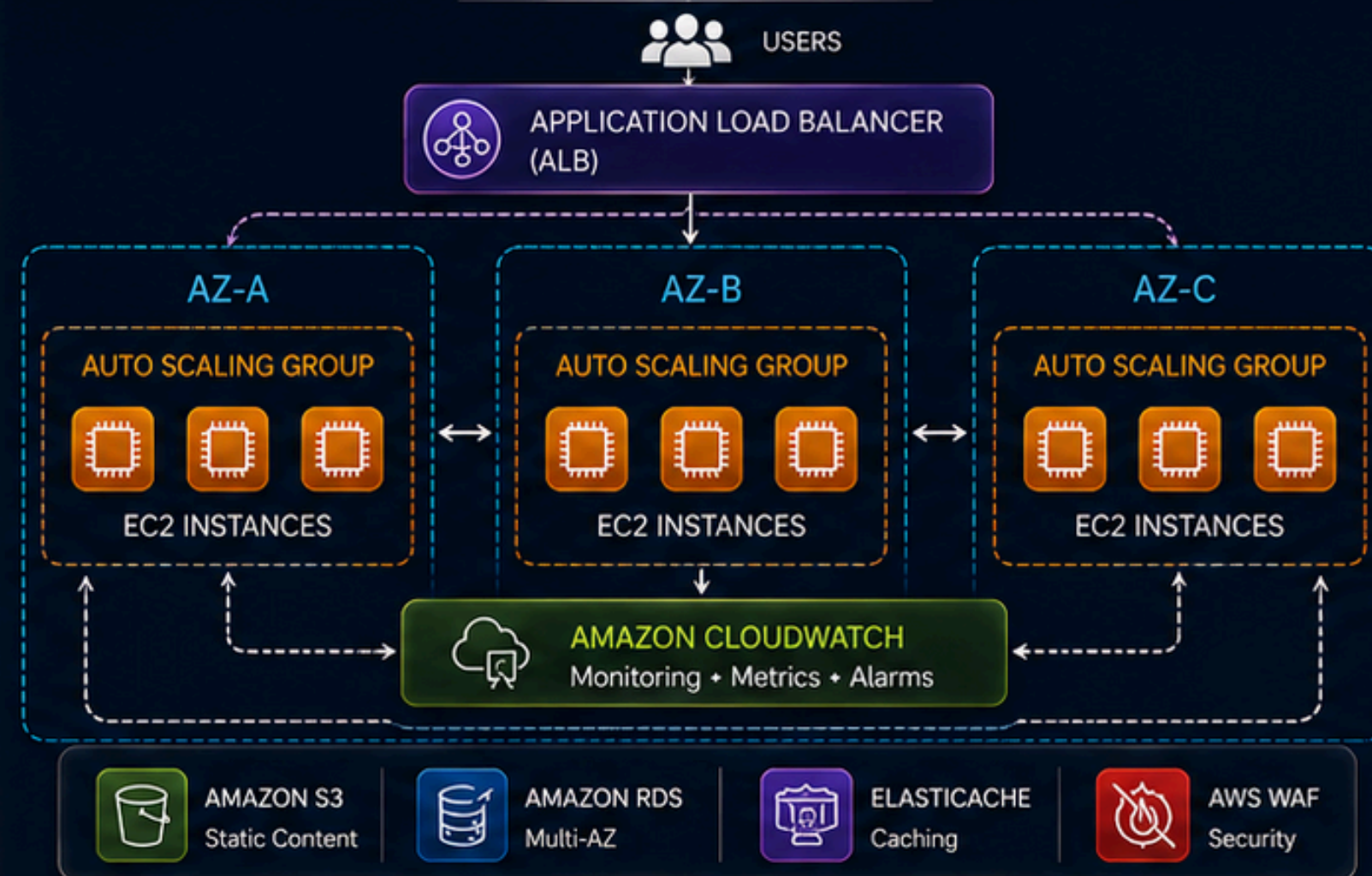
Application Load Balancer



Auto Scaling Group



Multiple Availability Zones



WHY?

Provides:



HIGH AVAILABILITY

Distributes traffic across multiple AZs



FAULT TOLERANCE

If an AZ fails, traffic is automatically routed to healthy AZs



ELASTIC SCALING

Automatically scales in/out based on demand to handle traffic spikes



COST OPTIMIZATION

Pay only for what you use. Scale down when traffic normalizes.



EXAM TIP

AWS solutions that combine ALB + ASG + Multi-AZ are almost always the answer for high availability and scalability.

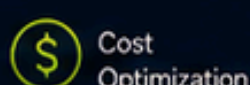
AWS WELL-ARCHITECTED PILLARS



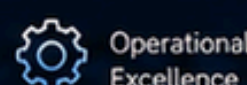
Reliability



Performance Efficiency



Cost Optimization



Operational Excellence



Security



KEY TAKEAWAY

Design for failure. Scale for demand. Deliver for customers.



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KEY TAKEAWAYS



EC2 INSTANCE TYPES

Choose based on workload.



HIGH AVAILABILITY

Use multiple Availability Zones.



FAULT TOLERANCE

Design for failure.



SCALING

Prefer Horizontal Scaling.



AUTO SCALING

Automates capacity management.

DESIGN • BUILD • SCALE • GOVERN
CLOUD EXCELLENCE DELIVERED GLOBALLY



“ Architects don’t build systems to survive success.
They build systems that scale because success is expected. ”



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MZ UCA CLOUD ACADEMY
Building Africa's Next
Generation of Cloud Architects



GOVERNANCE

Secure • Compliant • Trusted



RELIABILITY

Built for Business Continuity



SCALABILITY

Elastic • Resilient • Future-Ready



PERFORMANCE

Optimized for Excellence



COST OPTIMIZATION

Maximum Value • Minimum Waste